

CIS170 Programming Project 5

Wibaux University wants a summary report of its student population that shows how many in-district, out-of-district, and international students are at the freshman, sophomore, junior, senior, and graduate levels. The input file for this program is the school's student master file, WUSTDP:

```

A                               UNIQUE
A      R STUDREC
A      STUSSN      9S 0      TEXT('Student SSN')
A      SLNAME      15      TEXT('Last Name')
A      SFNAME      10      TEXT('First Name')
A      SMNAME      10      TEXT('Middle Name')
A      STREET      20      TEXT('Street Address')
A      CITY        15      TEXT('City')
A      STATE        2      TEXT('State')
A      ZIP          9S 0      TEXT('Zip+4')
A      PHONE       10S 0      TEXT('Phone')
A      CRDTOT       3S 0      TEXT('Tot. Earned Credits')
A      DCODE        1      TEXT('District Code')
A      ADMDAT       8S 0      TEXT('Date Admitted')
A      CLASS        1      TEXT('Classification')
A      GRDDAT       8      TEXT('Date Graduated')
A      SDEPT        3      TEXT('Dept. of Major')
A      GPA          3S 2      TEXT('Grade Point Avg.')
A      DEGREE       3      TEXT('Degree granted')
A      SEMAIL       30      TEXT('Student Email')
A      K STUSSN

```

Printchart:

Untitled / GTOTAL						
.....10.....20.....30.....40.....50.....60.....70.....80						
MM/DD/YYYY	Wibaux University				Page	1
	Student Population Summary Report					
	Student Residency					
Classification	In-District	Out-Of-District	International	Total		
Freshman	6,666	6,666	6,666	66,666		
Sophomores	6,666	6,666	6,666	66,666		
Juniors	6,666	6,666	6,666	66,666		
Seniors	6,666	6,666	6,666	66,666		
Total	66,666	66,666	66,666	666,666		

Note the following:

The district code is a code field where I = in-district, O = out-of-district, and F = international student status.

The classification field differentiates G (graduate) from U (undergraduate) students. ONLY UNDERGRADUATE STUDENTS ARE TO BE COUNTED.

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The school subdivides undergraduates based on earned credits: students with fewer than 30 credits are freshmen; those with 30–59 credits are sophomores; those with 60–89 credits are juniors; and those with 90 or more credits are seniors.

You will need to establish calculated fields for the desired counts as shown on the summary report printchart.

Note that this is a SUMMARY report, which means that all the output is printed AFTER the DoWhile loop ends, in other words, after all the data in the input file has been processed.

There are several ways to approach this problem. Certainly, decision-making logic is necessary. **SUBROUTINES are required in your solution** (Note that the video mentions subroutines as an option, but for the purposes of this project, SUBROUTINES are REQUIRED.) Try to make your logic as streamlined as possible, for the most efficient and easy to understand code. Because of the complexity of this program, the desired output results are shown below. If your numbers vary from this, you need to revisit your logic, which is of course, one of the many joys of programming!

Leave your last good compile and output report in your spooled files and email char.parker@muskegoncc.edu.

```
*...+...1...+...2...+...3...+...4...+...5...+...6...+...7...+...8.
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                               Student Population Summary Report
                               Student Residency
Classification  In-District  Out-Of-District  International  Total
-----
Freshman        9             5             5             19
Sophomores      1             0             0             1
Juniors         1             1             1             3
Seniors         1             0             1             2
Total          12            6             7            25
```